THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS TWO-CHAPTER TEST SET A

Chapters 1 & 2: **A Square and a Cube, Power Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a perfect square? 1

(a) 450 **(b)** 410 **(c)** 400 **(d)** 500

2. The value of 25^2 is: 1

(a) 625 **(b)** 600 **(c)** 50 **(d)** 650

3. $\sqrt{529}$ is equal to: 1

(a) 23 **(b)** 25 **(c)** 24 **(d)** 22

4. The sum of the first 8 odd natural numbers is: 1

(a) 72 **(b)** 64 **(c)** 63 **(d)** 16

5. How many non-square numbers lie between 7^2 and 8^2? 1

(a) 13 **(b)** 7 **(c)** 15 **(d)** 14

6. $5^2 \times 5^7$ equals: 1

(a) 5^9 **(b)** 25^9 **(c)** 5^{14} **(d)** 5^{-5}

7. $3^{10} \div 3^6$ equals: 1

(a) 3^{60} **(b)** 3^1 **(c)** 3^{16} **(d)** 3^4

8. $(2^3)^4$ equals: 1

(a) 2^{12} **(b)** 6^4 **(c)** 2^7 **(d)** 2^{81}

9. The value of 2^{-3} is: 1

(a) $\frac{1}{6}$ **(b)** -8 **(c)** $\frac{1}{8}$ **(d)** 8

10. The value of 4^{-2} is: 1

(a) 16 **(b)** $\frac{1}{8}$ **(c)** -16 **(d)** $\frac{1}{16}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Without calculating, say why 2468 cannot be a perfect square. 2

12. Find the cube root of 2744 by prime factorisation. 2

13. Why is any number raised to the power 0 equal to 1 (for non-zero numbers)? 2

14. Which is greater, 2^{10} or 10^2? Why? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A square garden has area 1296 m². Find its side. How much fencing is needed for the boundary? **3**

16. Express 0.0000327 in scientific notation and write the number of zeros after the decimal point before the first digit. **3**

17. A cubical tank holds 13824 litres (1 m³ = 1000 L). Find the length of its edge. **3**

OR

Show that the sum of two consecutive triangular numbers is a perfect square. Verify for 10 and 15. **3**

18. The distance of the Sun from the Earth is about 150000000 km. Write it in scientific notation and in terms of lakh/crore. **3**

OR

Simplify $\left(\frac{3}{4}\right)^{-2}$. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. (i) Find the square of 995 using $(1000 - 5)^2$. **5**
(ii) Find 97^2 using $(100 - 3)^2$.

OR

(i) State three properties of square numbers. **5**
(ii) Verify each with an example.

20. (i) Prove $a^m \times a^{-m} = 1$. **5**
(ii) Verify for $a = 3,\ m = 2$.

OR

(i) Write $\left(\frac{1}{2}\right)^{-5}$ without a negative exponent. **5**
(ii) Simplify $\dfrac{(-2)^3 \times (-2)^{-5}}{(-2)^{-1}}$.

END OF QUESTION PAPER — Total: 40 Marks

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